

Formality Tagging for Training English-to-Korean Neural Machine Translation

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Abstract

This project analyzes the impact of style-tagging on English-to-Korean Neural Machine Translation (NMT) models. Three models with different degrees of tags were trained with identical parameters and compared. Translations were evaluated using BLEU, chrF, and COMET metrics, as well as human judgment on content preservation and formality accuracy. The binary model (divided into formal vs. other) performed better than a tag-free model but only slightly. The multi-class tagging model had the highest BLEU and chrF scores indicating better word choice and fidelity to the input style. However, it did not have better COMET or human evaluated scores. Though tagging did successfully maintain the level of formality, it came at the cost of preserving semantics.

1. Introduction

Motivation: The Korean language uses different forms when conjugating verbs depending on the context of the situation. For example, when speaking in a formal setting or while talking to an elderly individual, statements will always contain the verb ending "습니다." When speaking casually, sentences often end in "해요/아/어". This distinction can make training and evaluating translation models difficult. A neural network may correctly translate a given English sentence, but the level of formality might differ from the training data. This can cause errors in minimizing loss as different verb endings may be seen as different words and penalized.

Objectives: To avoid these issues while training, we show the effectiveness of specifying the level of formality while training an English-to-Korean Neural Machine Translation Model. We compare three models with the exact same hyperparameters. The only difference is in the training data itself. One model will have a formal vs. non-formal tag that splits the training data into two categories. The other model will have 5 levels of formality and nuance, including a category for archaic and scriptural conjugations. The final model will have no such tags and serve as a control. Using standard metrics like BLEU, chrF and COMET, we expect to see improvements where the formality was specified within the datasets.

2. Related Work

There are several recent works that have addressed similar and adjacent issues to those tackled in this paper. For example, Chanung and Heeyoul [2019] is a paper about switching between formality using the word embedding space. They address this issue in the tokenization process rather than in training. This method found slight improvement in translation quality.

Zhang et al. [2018] discusses training with style transfer. This is not about forms of honorifics in Korean but rather styles of writing within the same language. It was an influential paper in encapsulating the style of a text when limited parallel data is present.

Jun-Nyeong et al. [2021] created a model that would input a sentence in high form and return the same sentence in low form. Vice versa could also

be completed. This is very similar to what we aim to do but with English input. Their training yielded success in almost perfectly transferring between the different forms.

Cho et al. [2022] was a non-neural project to build a corpus of parallel Korean data. They created a scheme to bulk up a corpus by creating duplicate sentences, simply written in another form. Their project only focused on changing questions or commands within a corpus. While all of these papers have inventive and successful ways to transfer styles, none have a solution as simple as tags within training data. This project aims to show adding tags is sufficient in improving formality within English-to-Korean translation.

3. Dataset

For this project, the church’s multilingual Korean-English data was used. It consists of General Conference talks, seminary and institute manuals, and various excerpts from Church magazines and websites. Each pair has an aligned English translation.

- From the Church’s data, 100,000 English-Korean pairs were randomly sampled.
Training set - 95,000
Validation set - 2,500
Test set - 2,500
- Using regular expressions, each pair had a formality tag placed at the beginning (see Experiments section for details).

4. Methodology

Neural Machine Translation

- Transformer using an Adam Optimizer with specified learning rate and warm up period. Validation accuracy and perplexity measurements were monitored during training.
- Each input to the model was a source sentence in English, optionally prefixed with a linguistic tag (e.g., <HIGH>, <LOW>) to condition the output style. These sentences were tokenized using a joint SentencePiece model with a vocabulary size of 8,000.

- The output was a Korean translation of the input sentence, generated by the model. Outputs were detokenized before evaluation.

5. Experiments

5.1. Baseline Model

The baseline system was built using the OpenNMT-py framework with a standard Transformer architecture. It was trained on a parallel English–Korean corpus without any form of linguistic tagging. The model served as a control for evaluating the impact of stylistic tagging on translation quality.

5.2. Tag-Enhanced Training Data

Two additional models were developed using tagged source sentences:

- **Binary Tags Model:** Each source sentence was prefixed with a binary tag (<HIGH> or <OTHER>) representing a simple stylistic label.
- **Multi-class Tags Model:** Each sentence was prefixed with one of five fine-grained stylistic tags: <HIGH>, <MIDDLE>, <LOW>, <ARCHAIC>, or <NONE>.

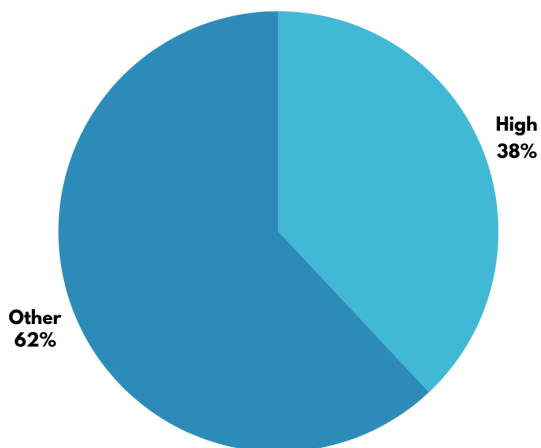
These tags were treated as standard tokens and included in the tokenization and training processes.

5.3. Tokenization and Training

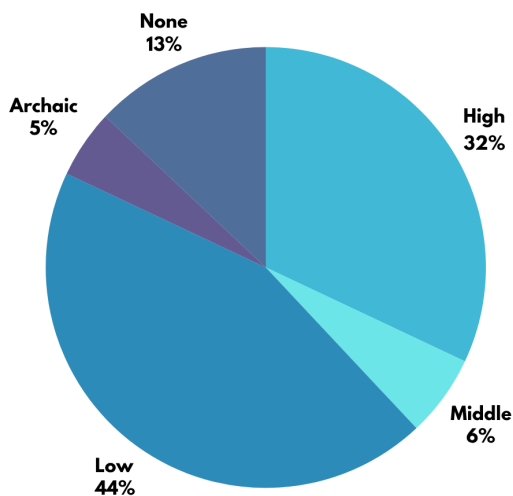
All data was preprocessed using SentencePiece with a joint BPE vocabulary of 8,000 tokens, trained on the combined English and Korean training corpora. This ensured consistent subword segmentation across both languages and improved handling of rare and morphologically rich words.

The models were trained in Google Colab using NVIDIA GPU. They used an Adam Optimizer with a learning rate of 0.005. All models had 4 attention heads with a 512-dimension hidden states. Training ran for 20,000 steps and took slightly more than 1 hour to finish. The only difference between these three models was their training data.

Binary Labelling



Multi-class Labelling



5.4. Evaluation Metrics

In order to measure the success of these models, we will use 3 different scoring methods. BLEU and chrF will measure the accuracy of word choice and directly compare it to the true translations. This will be essential in measuring whether or not a model got the level of formality correct. COMET scoring will also be measured. This will balance out the other scores as it puts less weight on word usage and more on preserving semantics within translation. This will ensure the content of the sentence

remains unchanged.

We will also use human evaluation on a small subset of outputs. This will be used to verify the results of scores and to see if meaning and fluency are at an acceptable level.

6. Results and Discussion

6.1. Quantitative Results

Model	BLEU	chrF	COMET
Baseline	15.44	28.0	0.7315
Binary Tags	18.05	30.1	0.7160
Multi-class Tags	21.04	32.2	0.6883

Table 1: Comparison of translation quality across metrics.

6.2. Analysis

When comparing scores between models, BLEU and chrF both show clear improvements from the baseline to the binary-tagged and multi-class-tagged models. These gains show us that the tagged models produce outputs that more closely resemble the reference translations at the word level. It has accurately matched the level of formality and form by seeing these tags. Both of these models seemed to work well in both of these metrics. It was exciting to see a steady increase in performance as the number of formality tags increased.

It was surprising to see the COMET score worked well for the baseline. It was anticipated that all three metrics would perform the best in the multi-class tagged model. The decrease in COMET scores highlights a serious question. Are we sacrificing semantic accuracy by focusing on word choice by using formality tags? Perhaps there is a way we could keep the COMET scores constant while also seeing the improvements in BLEU and chrF scores present in my results.

6.3. Human Evaluation

Native Korean speakers with fluency in English were tasked to rank samples of 35 sentences from

each model. The input of the model (English sentence with formality tag) was displayed along with three sample translations. Users award 1 to the best model and 3 to the worst. They were allowed to choose ties for any ranking. They also were asked to state whether the form was correct (even if the translation quality was poor). The results from these evaluations are as shown.

Model	Avg. Rank	Form Accuracy
Baseline	1.77	63.36%
Binary Tags	2.11	51.76%
Multi-class Tags	2.05	93.52%

Table 2: Human Evaluation: Content Ranking and Formality Accuracy for Each Model

The content rankings aligned well with the COMET scoring. This affirms our idea that while word choice is better, the meaning behind the translation suffered with tagging. It is interesting that the Baseline Model performed better than the Binary Model in getting the formality correct. However, neither model did as well as the Multi-class Model. This by far outperformed the other two, with over a 90% accuracy in choosing the correct verb form.

6.4. Qualitative Example

- Input: <LOW> Find out what the new youth curriculum means for you.
- Baseline: 새로운 교과 과정을 통해 여러분에게 어떤 의미가 있는지 알아보십시오.

Translation: *Through the new curriculum, find out what it means to you.*

- Binary: 새로운 청소년 교과 과정은 여러분에게 무엇을 의미하는지 알아본다.

Translation: *Explore what the new youth curriculum means for you.*

- Multi-class: 새로운 청소년 교과 과정이 자신에게 어떤 의미를 갖는지 알아본다.

Translation: *Explore what the new youth curriculum means for yourself.*

- Reference: 새로운 청소년 교과 과정이 여러분에게 어떤 의미를 갖는지 알아본다.

It is clear to see the Binary and Multi-class Models correctly got the low form correct in this example. There is some minor differences in word choice but both models produced output extremely similar to the reference. There are many examples where the content of the translation is missed entirely and is not similar to the reference. However, as seen by the scores and human evaluations these are not the majority.

7. Conclusion and Future Work

Results showed that both binary and multi-class tagging improved surface-level metrics such as BLEU and chrF. This means better alignment with reference translations in terms of word and character overlap. However, these gains came at the cost of lower COMET scores, suggesting a trade-off between surface similarity and semantic accuracy. While stylistic prompts can guide machine translation to more correct results, it may reduce fluency or faithfulness to the input text. The baseline model often got the level of formality wrong. Without the tags to indicate which form to conjugate in, it was merely a guessing game. This study shows the balance of form and meaning as both are crucial in translation. In future work, refining the tagging process could be a focus. Training on larger and better tagged data could improve the results found here. Additionally it would be interesting to see if an even more specific tagging system would improve the conjugations further. Balancing these two models could achieve a translation that is better equipped to translate in any given situation.

References

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